

Pursuing a **Minor** degree in **Artificial Intelligence and Data Science** at CMInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured an **All India Rank** of **635** in **Joint Entrance Exam (JEE) Advanced** among 0.15M+ candidates [Oct'21]
- Secured an **All India Rank** of **1323** in **Joint Entrance Exam (JEE) Mains** among 1M+ candidates [Sep'21]
- Recipient of the Kishore Vaigyanik Protsahan Yojana (**KVPY**) **Fellowship**, awarded by **IISc Bangalore** [Jan'21]
- Awarded **Best Project** in **Electronics Device Lab** for EEG measuring device, out of **200+** students [Apr'24]

PROFESSIONAL & RESEARCH EXPERIENCE

AI Agent Intern | **Attentions.AI**

[Sep'24 - Jan'25]

*Awarded a **Letter of Recommendation** in recognition of exceptional technical contributions*

- Developed a scalable automation framework using **PyAutoGUI** and **OmniParser** for dynamic web UI interaction
- Leveraged **Phi-3 Instruct** with **few-shot prompting** to generate dynamic action plans for complex workflows
- Achieved **80%** task completion accuracy by integrating **InternVL2 VLM** for visual state validation with retry logic

Text-to-Speech Development Intern | **Attentions.AI**

[May'24 - Jul'24]

To improve speech naturalness and real-time efficiency in Text-to-Speech systems

- Benchmarked TTS models and optimized the pipeline with **text chunking**, reducing inference latency by **50%**
- Enhanced speech expressiveness via **emotion-conditioned attention** mechanism, improving MOS by over **10%**

Multi Document Multi Modal Reasoning | **Thesis Project**

[Jan'25 - Present]

Dual Degree Project-1 | Guide: Prof. Ganesh Ramakrishnan, Department of Computer Science and Engineering

- Improved large-scale image retrieval via **submodular subset selection**, boosting performance in 10k+ corpora
- Enhanced query robustness via **data augmentation & context similarity optimization**, improving retrieval by 5%
- Extending to multi-document, 1k+ files using **ensemble-similarity**, LLM-filtering, and Inverted file indexing

Singer Classification using Gesture Features | **Research Project**

[Aug'24 - Nov'24]

Supervised Research Exposition | Guide: Prof. Preeti Rao, Department of Electrical Engineering

- Implemented a multi-class classification model for Hindustani singers with **88.28%** accuracy using gesture data
- Extracted and analyzed features like **LDJ**, **SPARC** from 3D motion capture data to quantify motion smoothness
- Built an end-to-end pipeline for processing and classifying time-series gesture data using **SVM** with **RBF** kernel

KEY PROJECTS

Bitcoin Trading Strategy | **Summer Project**

[Jun'25 - Jul'25]

Summer of Quant | Quant Community, IIT Bombay

- Completed a program on Financial Modeling and Time Series Analysis, covering **Derivative Pricing** and Hedging
- Designed strategies with momentum and **breakout reversal** logic, using long-short entries & trailing-stop exits
- Achieved Sharpe ratios up to **2.21**, net profits up to **2.5x**, controlled drawdowns, verified against **lookahead bias**

Limestone Data Challenge | **Algorithmic Trading Hackathon**

[Mar'23 - Apr'23]

Organised by Finance & Analytics Club, IIT Bombay in association with TOWER Research Capital

- Clustered **100 stocks** into **4** distinct sectors using correlation based analysis of returns between stocks and indices
- Predicted sector index returns with **regression models**, achieving up to **R = 0.49 correlation** with the actual prices
- Designed a covariance driven **long-short trading strategy** with net-zero exposure, achieving a **Sharpe ratio** of **1.7**

Stock Price Prediction using LSTM | **Course Project**

[Aug'23 - Sep'23]

EE782 - Advanced Topics in Machine Learning | Guide: Prof. Amit Sethi

- Developed a **PyTorch LSTM** module along with a flexible data loader to enable dynamic stock price prediction
- Designed a trading strategy class with **EMA indicator** to automate stock trades based on LSTM's predictions
- Achieved a **21%** profit with LSTM's predictions on test data compared to **12%** profit from buy-and-hold strategy

Understanding Trading of Currencies and Commodities in Stock Market

[Jun'22 - Aug'22]

Finsearch | Finance Club, IIT Bombay

- Secured a place in the **Top 15** out of 240+ teams in a 4-member team, analyzing the **Forex** and **commodity markets**
- Researched major factors influencing short/long-term market trends, formulating data-driven **trading strategies**
- Developed and backtested a model for commodity and Forex trading to validate the performance and accuracy

OTHER PROJECTS

AI-Generated Image Classification and Artifact Detection | *Industrial Project* [Nov'24 - Dec'24]

Inter-IIT Tech Meet 13.0 | Adobe Research Challenge

- Designed pipeline with **DenseNet121** & **MobileNetV3** for AI-vs-Real image classification, achieving **98%** accuracy
- Fine-tuned Qwen2-VL with **QLoRA** and **Tree of Thoughts** to detect 70+ artifacts, achieving an F1-score of **0.79**
- Incorporated **adversarial defenses** (FGSM, PGD), enabling cross-model generalization and CPU inference of **<2s**

IITB Q&A Smartbot | *Course Project*

[Sep'23 - Nov'23]

EE782 - Advanced Topics in Machine Learning | Guide: Prof. Amit Sethi

- Scraped **20+** IITB websites using **BeautifulSoup** and Requests to build a domain-specific knowledge database
- Built **Agentic RAG** chatbot using **LangGraph** and **ChromaDB** storage with dynamic routing for accurate response
- Integrated **LLaMA2-13B-GPTQ** model with **LangSmith** for real-time evaluation, output moderation, and tracing

Image Captioning | *Summer Project*

[May'23 - Jul'23]

Seasons of Code | Web and Coding Club, IIT Bombay

- Implemented an algorithm on Neural Image Caption Generator by Google using **CNN** as the image descriptor
- Used pretrained **VGGNet16** for feature extraction and built a preprocessing pipeline on the Flickr8k dataset
- Predicted caption sequences with **LSTM**, addressing vanishing gradients and achieving a **BLEU-4 score** of **0.1436**

Denoising of Synthetically Noised Images | *Course Project*

[Jan'24 - Apr'24]

GNR638 - Remote Sensing and Machine Learning | Guide: Prof. Biplab Banerjee

- Generated blurred image pairs using 3 **Gaussian filters** to create diverse synthetic data for robust model training
- Trained a **GAN-based** deblurring model with **VGG perceptual** and **Wasserstein loss**, boosting PSNR by **+3.8 dB**
- Achieved an average **25.69 dB** PSNR on synthetic set using a **ResNet-based generator** for sharp image restoration

Developing Face Recognition and Generation Models | *Course Project*

[Sep'23 - Oct'23]

EE782 - Advanced Topics in Machine Learning | Guide: Prof. Amit Sethi

- Constructed **Siamese network** using **ResNet18**, exploring diverse configurations of dropout, LRs, and optimizers
- Built a validation set for predicting whether a pair of images belong to the same person, achieving **75%** accuracy
- Implemented **DC-GAN** architecture with **BCE loss**, leveraging the LFW dataset to generate synthetic face images

Detecting Depression through Tweets | *Self Project*

[Dec'22 - Jan'23]

Detecting signs of depression from tweets using sentiment analysis and deep learning

- Preprocessed tweet data using tokenization, stopword removal, stemming, lemmatization for sentiment analysis
- Developed a two-layer **LSTM** for sentiment prediction, achieving **82%** accuracy vs 50% with **Logistic Regression**

POSITION OF RESPONSIBILITY

WiDS Project Mentor | *Analytics Club, IIT Bombay*

[Dec'24 - Jan'25]

- Mentored **10** students on Bokeh effect, applying **DeepLab** for segmentation and **OpenCV** for background blur
- Curated a project after exploring multiple ideas to enhance learning in object segmentation and image processing

Teaching Assistant | *EE, IIT Bombay*

[Aug'25 - Present]

EE353 - Introduction to Data Science and Machine Learning | Guide: Prof. Jayakrishnan Nair

- Coordinating in a group of **7** TAs; responsible for the assessments, grading and counseling of **100+** course students
- Assisting professor in moderating classes, proctoring examinations, catering to students' course-related queries

TECHNICAL SKILLS

- Programming Languages:** C++, Python, SQL, JavaScript, VHDL, Embedded C, HTML, CSS, $\text{\LaTeX}$
- Machine Learning:** NumPy, Pandas, TensorFlow, Keras, PyTorch, OpenCV, Scikit-Learn, NLTK
- Software Tools:** GitHub, Matlab, Selenium, Excel, ArduinoIDE, Intel Quartus, GNU-Radio

COURSES UNDERTAKEN

Data Science and AI

Advanced Topics in Machine Learning, Advanced Topics in AI Data and Policy, **Optimization in ML**, Image Processing, Computer Programming and Utilization

Maths and Physics

Applied Linear Algebra, Complex Analysis, Differential Equations, Calculus I & II, **Probability and Random Processes**, Digital Signal Processing, Communication Systems, **Markov Chains**, Number Theory and Cryptography, Quantum Physics

EXTRACURRICULAR ACTIVITIES

Technical	- Achieved Gold Level in the WorldQuant BRAIN for excellence in quantitative modeling (2024) - Successfully completed 8-week Data Structures and Algorithms Bootcamp by GFG (2025) - Completed Game Theory and Tinkering Bootcamp under Learner Space, IIT Bombay (2022) - Worked in a team to build an RC trainer aircraft organised by Aeromodelling Club (2022)
Cultural and Sports	Stood 1st in Gyrations, Inter-Hostel Dance General Championship of IIT Bombay (2024) - Participated in Annual InSync Dance Show with footfall of 1.5k+, by InSync Club (2022) - Participated in Intra-Department Badminton Tournament and reached the top-8 teams (2022)